EXERCISE USING CONTROL STRUCTURES

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Aim:

Write a **Java code** to read the customer details, previous reading and current reading and connect type. Print the **electricity bill** for the total amount of units consumed as per the following criteria:

Consumer Type	Usage Slab (Units)	Rate per Unit (in Rs.)
Domestic	Up to 100	4.80
	101 – 200	5.80
	Above 200	6.50
Non-Domestic	Up to 100	6.05

Source Code:

```
import java.io.*;

import java.lang.*;

public class ele {

    public static void main(String args[]) throws IOException {

        DataInputStream d = new DataInputStream(System.in);

        double total = 0;

        System.out.println("Enter the Previous Reading:");

        int pre = Integer.parseInt(d.readLine());

        System.out.println("Enter the Present Reading:");
```

```
int present = Integer.parseInt(d.readLine());
// BUG FIX 1: Present minus Previous
int unit = present - pre;
System.out.println("Enter the Consumer Type (d for Domestic):");
String type = d.readLine();
System.out.println("-----------------------");
System.out.println("The Usage Slab is :" + unit);
if (type == 'd') {
    if (unit <= 100) {
        System.out.println("Rate per Unit: RS.4.80");
        total = unit * 4.80;
    }
    else if (unit <= 200) {
        // BUG FIX 3: Cumulative Calculation
        System.out.println("Rate per Unit: RS.5.80");
        total = (100 * 4.80) + (unit - 100) * 5.80;
    }
    else {
         System.out.println("Rate per Unit: RS.6.50");
        total = (100 * 4.80) + (100 * 5.80) + (unit - 200) * 6.50;
    }
} else {
    // Non-Domestic logic
     System.out.println("Rate per Unit: RS.6.05");
```

```
            total = unit * 6.05;

        }

        System.out.println("The Bill Amount :" + total);

    }

}
```

Output :

```
D:\java prgm>java ele
Enter the Previous Reading:
2800
Enter the Present Reading:
3000
Enter the Consumer Type (d for Domestic):
d
-----------------------
The Usage Slab is :200
Rate per Unit: RS.5.80
The Bill Amount :1060.0
```

```
D:\java prgm>java ele
Enter the Previous Reading:
2500
Enter the Present Reading:
3000
Enter the Consumer Type (d for Domestic):
nd
-----------------------
The Usage Slab is :500
Rate per Unit: RS.6.05
The Bill Amount :3025.0
```

Result:

Thus, The java program using control structures to calculate the electricity bill based on consumer type was executed successfully .